OpenPBS Installation & Configuration Guide

PBS Server: hpc-master (192.168.100.1)

PBS Execution Nodes: compute1 (192.168.100.13), compute2 (192.168.100.14)

PBS Client / Submit Host: login (192.168.100.11)

OpenPBS Version: 23.06.06

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1. What is OpenPBS?

OpenPBS (Portable Batch System) is an open-source workload management and job scheduling system designed for HPC (High Performance Computing) clusters. It manages the submission, scheduling, monitoring, and execution of computational jobs across multiple nodes in a cluster.

OpenPBS was originally developed by NASA and is now maintained by Altair Engineering as an open-source project. It is one of the most widely used job schedulers in HPC environments worldwide.

In simple terms, OpenPBS acts as a **traffic controller** for your cluster — users submit jobs, PBS decides when and where to run them based on available resources, and automatically distributes work across compute nodes.

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2. Concepts

2.1 Job

A job is a unit of work submitted to PBS. It is typically a shell script containing the commands to execute along with resource requirements such as number of CPUs, memory, and walltime.

2.2 Queue

A queue holds submitted jobs before they are scheduled for execution. Jobs wait in the queue until the required resources become available on compute nodes. In this cluster the default queue is named workq.

2.3 Scheduler

The PBS scheduler (pbs_sched) continuously monitors available resources and pending jobs. It decides which job runs next based on resource availability, job priority, and scheduling policies.

2.4 PBS Mom (Machine Oriented Mini-server)

PBS Mom (pbs_mom) runs on each execution node. It is responsible for actually executing jobs, monitoring their resource usage, and reporting status back to the PBS server. Without pbs_mom running on a compute node, that node cannot accept or run any jobs.

2.5 PBS Comm

PBS Comm (pbs_comm) handles all communication between the PBS server and execution nodes. It acts as a message router within the PBS system.

2.6 Walltime

Walltime is the maximum real-world time a job is allowed to run. If a job exceeds its walltime, PBS automatically kills it. This prevents runaway jobs from blocking resources.

2.7 Select Statement

The select statement defines the resources a job needs. For example select=1:ncpus=2:mem=4gb means the job needs 1 node with 2 CPUs and 4GB of memory.

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3. PBS Architecture in This Cluster

```
hpc-master (PBS Server + Scheduler)
 ├── pbs_server   — accepts job submissions, manages job database
 ├── pbs_sched    — schedules jobs based on resources
 ├── pbs_comm     — handles inter-node communication
 └── PostgreSQL   — stores PBS job and node database
       |
       | Port 15001 (server)
       | Port 15002 (mom)
       | Port 17001 (comm)
       |
  ┌─────┴──────────────────┐      ┌──────────┐
  |              |     |     |
compute1       compute2     login
(pbs_mom)       (pbs_mom)   (PBS Client)
Execution Node     Execution   Job Submit
               Node       Host
```

PBS Component Summary:

Component	Node	Process	Purpose
PBS Server	hpc-master	pbs_server	Manages jobs and nodes
PBS Scheduler	hpc-master	pbs_sched	Schedules jobs
PBS Comm	hpc-master	pbs_comm	Routes messages
PBS Mom	compute1	pbs_mom	Executes jobs
PBS Mom	compute2	pbs_mom	Executes jobs

PBS Client	login	—	Job submission

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4. RPM Package Selection

OpenPBS 23.06.06 provides separate RPM packages for each role. The correct package was downloaded and SCP'd to each node:

Node	RPM Package	Purpose
hpc-master	openpbs-server-23.06.06-0.x86_64.rpm	PBS server, scheduler, comm
compute1	openpbs-execution-23.06.06-0.x86_64.rpm	PBS Mom execution daemon
compute2	openpbs-execution-23.06.06-0.x86_64.rpm	PBS Mom execution daemon
login	openpbs-client-23.06.06-0.x86_64.rpm	Job submission tools

Note: RPMs were downloaded from https://www.openpbs.org/download/ for Rocky Linux specifically on a local Windows machine and transferred to each node using SCP since the cluster nodes do not have external DNS resolution.

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5. Pre-Installation Requirement — Fix /etc/hosts

A critical prerequisite before installing PBS on any node is ensuring /etc/hosts has correct IP-to-hostname mappings. PBS Mom fails to start if the node's hostname resolves to 127.0.0.1 (localhost) instead of its real cluster IP.

The following /etc/hosts was applied to **all nodes** (hpc-master, compute1, compute2, login, storage):

127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4

::1 localhost localhost.localdomain localhost6 localhost6.localdomain6

Cluster nodes

192.168.100.1 hpc-master.hpc.local hpc-master

192.168.100.11 login.hpc.local login

192.168.100.13 compute1.hpc.local compute1

192.168.100.14 compute2.hpc.local compute2

192.168.100.12 storage.hpc.local storage

Important: Never map a node's own FQDN to 127.0.0.1. PBS Mom requires the node's real network IP to communicate with the PBS server.

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6. PBS Server Installation — hpc-master

All steps performed via direct SSH to hpc-master.

6.1 Transfer RPM to hpc-master

From Windows local machine

scp openpbs-server-23.06.06-0.x86_64.rpm root@192.168.179.140:/tmp/

#Downloaded from - https://www.openpbs.org/download/

6.2 Install PBS Server RPM

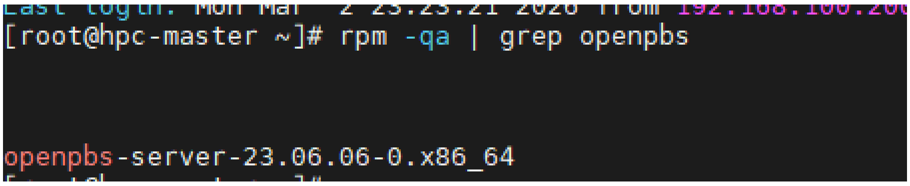
dnf install /tmp/openpbs-server-23.06.06-0.x86_64.rpm -y

Verify installation

rpm -qa | grep openpbs

Output:

openpbs-server-23.06.06-0.x86_64



6.3 Add PBS to System PATH

echo 'export PATH=$PATH:/opt/pbs/bin:/opt/pbs/sbin' >> /root/.bashrc

source /root/.bashrc

6.4 Start PBS Server

/etc/init.d/pbs start

systemctl enable pbs

Output:

Starting PBS

PBS comm — pbs_comm ready, Proxy Name: hpc-master.hpc.local:17001

PBS sched — Connecting to PBS dataservice...connected

PBS server — started

6.5 Verify PBS Processes

ps aux | grep pbs

Confirmed processes running:

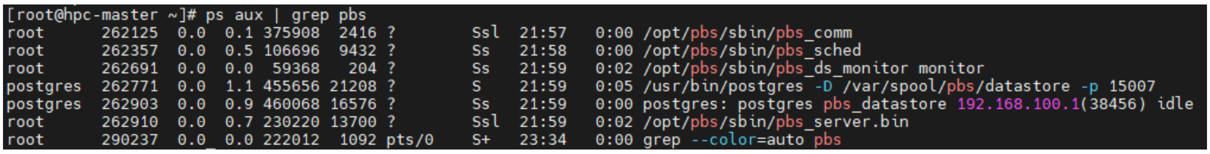
/opt/pbs/sbin/pbs_comm

/opt/pbs/sbin/pbs_sched

/opt/pbs/sbin/pbs_ds_monitor

/usr/bin/postgres (PBS datastore)

/opt/pbs/sbin/pbs_server.bin



```
[root@hpc-master ~]# ps aux | grep pbs
root      262125  0.0  0.1 375908  2416 ?        Ssl  21:57   0:00 /opt/pbs/sbin/pbs_comm
root      262357  0.0  0.5 106696  9432 ?        Ss   21:58   0:00 /opt/pbs/sbin/pbs_sched
root      262691  0.0  0.0  59368   204 ?        Ss   21:59   0:02 /opt/pbs/sbin/pbs_ds_monitor monitor
postgres  262771  0.0  1.1 455656 21208 ?        S    21:59   0:05 /usr/bin/postgres -D /var/spool/pbs/datastore -p 15007
postgres  262903  0.0  0.9 460068 16576 ?        Ss   21:59   0:00 postgres: postgres pbs_datastore 192.168.100.1(38456) idle
root      262910  0.0  0.7 230220 13700 ?        Ssl  21:59   0:02 /opt/pbs/sbin/pbs_server.bin
root      290237  0.0  0.0 222012  1092 pts/0    S+   23:34   0:00 grep --color=auto pbs
```

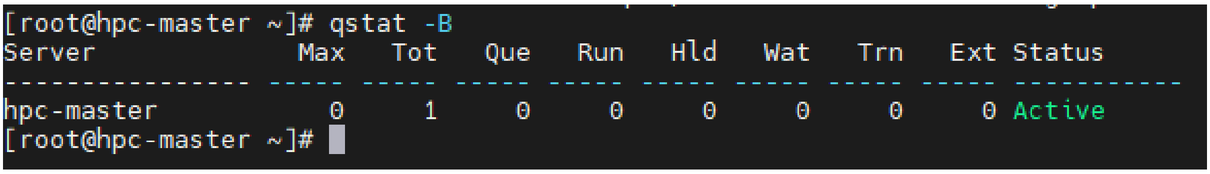
6.6 Verify PBS Server Status

qstat -B

Output:

Server   Max Tot Que Run Hld Wat Trn Ext Status

hpc-master   0   0   0   0   0   0   0   0   Active



```
[root@hpc-master ~]# qstat -B
Server             Max   Tot   Que   Run   Hld   Wat   Trn   Ext Status
---------------- ----- ----- ----- ----- ----- ----- ----- ----- -----------
hpc-master           0     1     0     0     0     0     0     0 Active
[root@hpc-master ~]#
```

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7. PBS Server Configuration — hpc-master

7.1 Configure Server Properties

qmgr -c "set server flatuid=true"

qmgr -c "set server acl_roots+=root@hpc-master.hpc.local"

qmgr -c "set server acl_roots+=root@*"

qmgr -c "set server managers+=root@hpc-master.hpc.local"

qmgr -c "set server operators+=root@hpc-master.hpc.local"

qmgr -c "set server scheduling=true"

qmgr -c "set server job_history_enable=true"

Explanation of server properties:

Property	**Value**	**Meaning**
flatuid	true	Maps user UIDs across nodes without requiring identical UIDs
acl_roots	root@*	Allows root to submit jobs from any host
scheduling	true	Enables the scheduler to run jobs
job_history_enable	true	Keeps history of completed jobs

7.2 Configure Default Queue

workq already existed from initial setup

qmgr -c "set queue workq queue_type=execution"

qmgr -c "set queue workq started=true"

qmgr -c "set queue workq enabled=true"

qmgr -c "set server default_queue=workq"

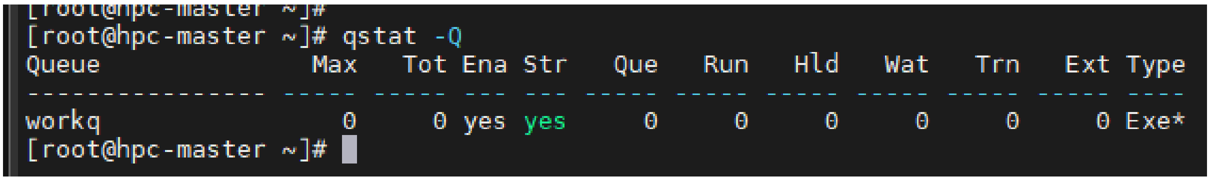
Verify queue:

qstat -Q

Output:

Queue Max Tot Ena Str Que Run Hld Wat Trn Ext Type

workq 0 0 yes yes 0 0 0 0 0 0 Exe



7.3 Add Compute Nodes

qmgr -c "create node compute1"

qmgr -c "create node compute2"

qmgr -c "set node compute1 resources_available.ncpus=2"

qmgr -c "set node compute2 resources_available.ncpus=2"

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8. PBS Execution Node Installation — compute1 & compute2

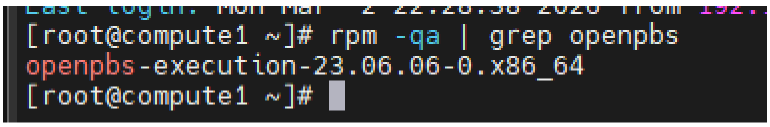
The same steps were applied on both compute nodes.

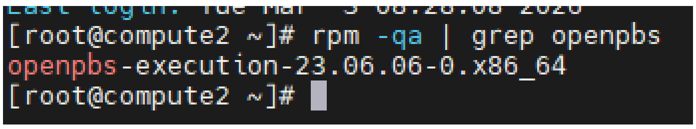
8.1 Install Execution RPM

ssh root@compute1

dnf install /tmp/openpbs-execution-23.06.06-0.x86_64.rpm -y

rpm -qa | grep openpbs





8.2 Configure /etc/pbs.conf

vi /etc/pbs.conf

Contents:

PBS_SERVER=hpc-master.hpc.local

PBS_START_SERVER=0

PBS_START_SCHED=0

PBS_START_COMM=0

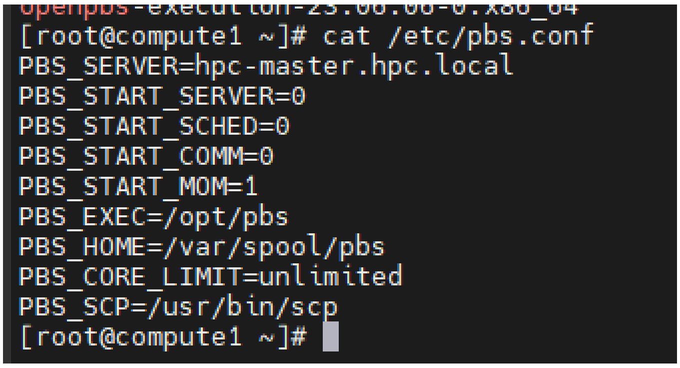
PBS_START_MOM=1

PBS_EXEC=/opt/pbs

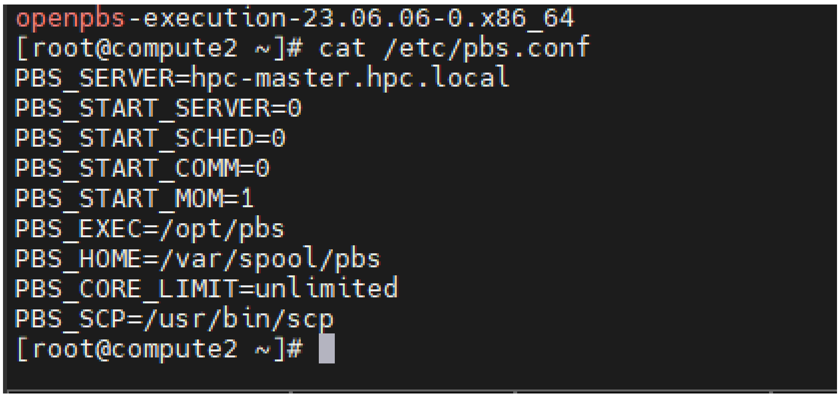
PBS_HOME=/var/spool/pbs

PBS_CORE_LIMIT=unlimited

PBS_SCP=/usr/bin/scp



```
[root@compute1 ~]# cat /etc/pbs.conf
PBS_SERVER=hpc-master.hpc.local
PBS_START_SERVER=0
PBS_START_SCHED=0
PBS_START_COMM=0
PBS_START_MOM=1
PBS_EXEC=/opt/pbs
PBS_HOME=/var/spool/pbs
PBS_CORE_LIMIT=unlimited
PBS_SCP=/usr/bin/scp
[root@compute1 ~]#
```



```
openpbs-execution-23.06.06-0.x86_64
[root@compute2 ~]# cat /etc/pbs.conf
PBS_SERVER=hpc-master.hpc.local
PBS_START_SERVER=0
PBS_START_SCHED=0
PBS_START_COMM=0
PBS_START_MOM=1
PBS_EXEC=/opt/pbs
PBS_HOME=/var/spool/pbs
PBS_CORE_LIMIT=unlimited
PBS_SCP=/usr/bin/scp
[root@compute2 ~]#
```

Explanation of pbs.conf settings:

Setting	Value	Meaning
PBS_SERVER	hpc-master.hpc.local	FQDN of the PBS server
PBS_START_MOM	1	Start pbs_mom on this node
PBS_START_SERVER	0	Do not start PBS server here

PBS_START_SCHED	0	Do not start scheduler here
PBS_START_COMM	0	Do not start comm here

8.3 Start PBS Mom

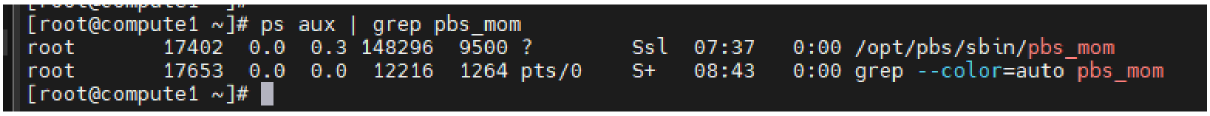
/opt/pbs/sbin/pbs_mom

Verify

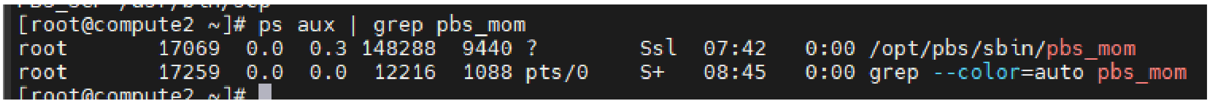
ps aux | grep pbs_mom | grep -v grep

Output:

root 17402 0.7 0.2 148400 8848 ? Ssl pbs_mom running



```
[root@compute1 ~]# ps aux | grep pbs_mom
root      17402  0.0  0.3 148296  9500 ?        Ssl  07:37   0:00 /opt/pbs/sbin/pbs_mom
root      17653  0.0  0.0  12216  1264 pts/0    S+   08:43   0:00 grep --color=auto pbs_mom
[root@compute1 ~]#
```



```
[root@compute2 ~]# ps aux | grep pbs_mom
root      17069  0.0  0.3 148288  9440 ?        Ssl  07:42   0:00 /opt/pbs/sbin/pbs_mom
root      17259  0.0  0.0  12216  1088 pts/0    S+   08:45   0:00 grep --color=auto pbs_mom
[root@compute2 ~]#
```

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9. PBS Client Installation — login node

9.1 Install Client RPM

ssh root@login

dnf install /tmp/openpbs-client-23.06.06-0.x86_64.rpm -y

9.2 Configure /etc/pbs.conf

vi /etc/pbs.conf

Contents:

PBS_SERVER=hpc-master.hpc.local

PBS_START_SERVER=0

PBS_START_SCHED=0

PBS_START_COMM=0

PBS_START_MOM=0

PBS_EXEC=/opt/pbs

PBS_HOME=/var/spool/pbs

PBS_CORE_LIMIT=unlimited

PBS_SCP=/usr/bin/scp

```
[root@login ~]# cat /etc/pbs.conf
PBS_SERVER=hpc-master.hpc.local
PBS_START_SERVER=0
PBS_START_SCHED=0
PBS_START_COMM=0
PBS_START_MOM=0
PBS_EXEC=/opt/pbs
PBS_HOME=/var/spool/pbs
PBS_CORE_LIMIT=unlimited
PBS_SCP=/usr/bin/scp
[root@login ~]#
```

Note: PBS_START_MOM=0 on login node — it only submits jobs, does not execute them.

9.3 Add PBS to PATH

echo 'export PATH=$PATH:/opt/pbs/bin:/opt/pbs/sbin' >> /root/.bashrc

source /root/.bashrc

---

10. Verification

10.1 Check All Nodes are Free

pbsnodes -a

Output:

compute1

state = free

pcpus = 2

resources_available.mem = 3012244kb

resources_available.ncpus = 2

pbs_version = 23.06.06

compute2

state = free

pcpus = 2

resources_available.mem = 3012264kb

resources_available.ncpus = 2

pbs_version = 23.06.06

```
[root@hpc-master ~]# pbsnodes -a
compute1
     Mom = compute1.hpc.local
     Port = 15002
     pbs_version = 23.06.06
     ntype = PBS
     state = free
     pcpus = 2
     resources_available.arch = linux
     resources_available.host = compute1
     resources_available.mem = 3012244kb
     resources_available.ncpus = 2
     resources_available.vnode = compute1
     resources_assigned.accelerator_memory = 0kb
     resources_assigned.hbmem = 0kb
     resources_assigned.mem = 0kb
     resources_assigned.naccelerators = 0
     resources_assigned.ncpus = 0
     resources_assigned.vmem = 0kb
     resv_enable = True
     sharing = default_shared
     license = l
     last_state_change_time = Mon Mar  2 23:13:13 2026
     last_used_time = Mon Mar  2 22:52:37 2026

compute2
     Mom = compute2.hpc.local
     Port = 15002
     pbs_version = 23.06.06
     ntype = PBS
     state = free
     pcpus = 2
     resources_available.arch = linux
     resources_available.host = compute2
     resources_available.mem = 3012264kb
     resources_available.ncpus = 2
     resources_available.vnode = compute2
```

10.2 Check Server and Queue Status

qstat -B

qstat -Q

Output:

Server Status

hpc-master Active

Queue Ena Str Type

workq yes yes Execution

```
[root@hpc-master ~]# qstat -B
Server             Max   Tot   Que   Run   Hld   Wat   Trn   Ext  Status
---------------- ----- ----- ----- ----- ----- ----- ----- ----- -----------
hpc-master           0     1     0     0     0     0     0     0 Active
[root@hpc-master ~]#
```

```
hpc-master              0    1    0    0    0    0    0    0 Active
[root@hpc-master ~]# qstat -Q
Queue              Max   Tot Ena Str   Que   Run   Hld   Wat   Trn   Ext Type
---------------- ----- ----- --- --- ----- ----- ----- ----- ----- ----- ----
workq                0     0 yes yes     0     0     0     0     0     0 Exe*
[root@hpc-master ~]#
```

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11. Job Submission — PBS Directives Reference

When submitting jobs, PBS directives are added at the top of the job script using #PBS:

Directive	**Example**	**Meaning**
-N	#PBS -N MyJob	Job name
-l select	#PBS -l select=1:ncpus=2	Resources: 1 node, 2 CPUs
-l mem	#PBS -l select=1:mem=2gb	Request 2GB memory
-l walltime	#PBS -l walltime=01:00:00	Max runtime 1 hour
-q	#PBS -q workq	Submit to workq queue
-o	#PBS -o output.log	Redirect stdout
-e	#PBS -e error.log	Redirect stderr
-j oe	#PBS -j oe	Merge stdout and stderr

Sample Job Script

#!/bin/bash

#PBS -N TestJob

#PBS -l select=1:ncpus=1

#PBS -l walltime=00:00:30

#PBS -q workq

echo "=== Job Started ==="

echo "Running on host: $(hostname)"

echo "Date and Time: $(date)"

echo "Job ID: $PBS_JOBID"

echo "=== Job Finished ==="

```
test_job.sh
[root@login tmp]# cat test_job.sh
#!/bin/bash
#PBS -N TestJob
#PBS -l select=1:ncpus=1
#PBS -l walltime=00:00:30
#PBS -q workq

echo "=== Job Started ==="
echo "Running on host: $(hostname)"
echo "Date and Time: $(date)"
echo "CPUs allocated: $PBS_NUM_PPN"
echo "Job ID: $PBS_JOBID"
echo "=== Job Finished ==="
[root@login tmp]#
```

Submit:

cd /tmp

qsub /test_job.sh

Monitor:

qstat -a

```
[root@login tmp]# qstat -a
[root@login tmp]# qsub test_job.sh
4.hpc-master
[root@login tmp]# qstat -a

hpc-master:
                                                            Req'd  Req'd   Elap
Job ID          Username Queue    Jobname    SessID NDS TSK Memory Time  S Time
--------------- -------- -------- ---------- ------ --- --- ------ ----- - -----
4.hpc-master    root     workq    TestJob      --     1   1    --  00:00 E 00:00
[root@login tmp]# qstat -a
[root@login tmp]#
```

Job states:

State	Meaning
Q	Queued — waiting for resources
R	Running
F	Finished
H	Held
E	Exiting

12. Commands

Purpose	Command
Submit a job	qsub job.sh
Check job status	qstat -a
Check server status	qstat -B
Check queue status	qstat -Q
Check node status	pbsnodes -a
Delete a job	qdel <jobid>
Hold a job	qhold <jobid>
Release a held job	qrls <jobid>
View job details	qstat -f <jobid>
View completed jobs	qstat -x
Check PBS server config	qmgr -c "print server"
Start PBS	/etc/init.d/pbs start
Stop PBS	/etc/init.d/pbs stop
Start PBS mom manually	/opt/pbs/sbin/pbs_mom

```
[root@login tmp]#
[root@login tmp]# qstat -x
Job id            Name             User              Time Use S Queue
----------------  ---------------- ----------------  -------- - -----
1.hpc-master      TestJob          root              00:00:00 F workq
2.hpc-master      TestJob          root              00:00:00 F workq
3.hpc-master      TestJob          root              00:00:00 F workq
4.hpc-master      TestJob          root              00:00:00 F workq
[root@login tmp]#
```

```
[root@login tmp]#
[root@login tmp]# qstat -f 4
qstat: 4.hpc-master.hpc.local Job has finished, use -x or -H to obtain historical job information
[root@login tmp]#
```